



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VIJAYA AUTO SERVICE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

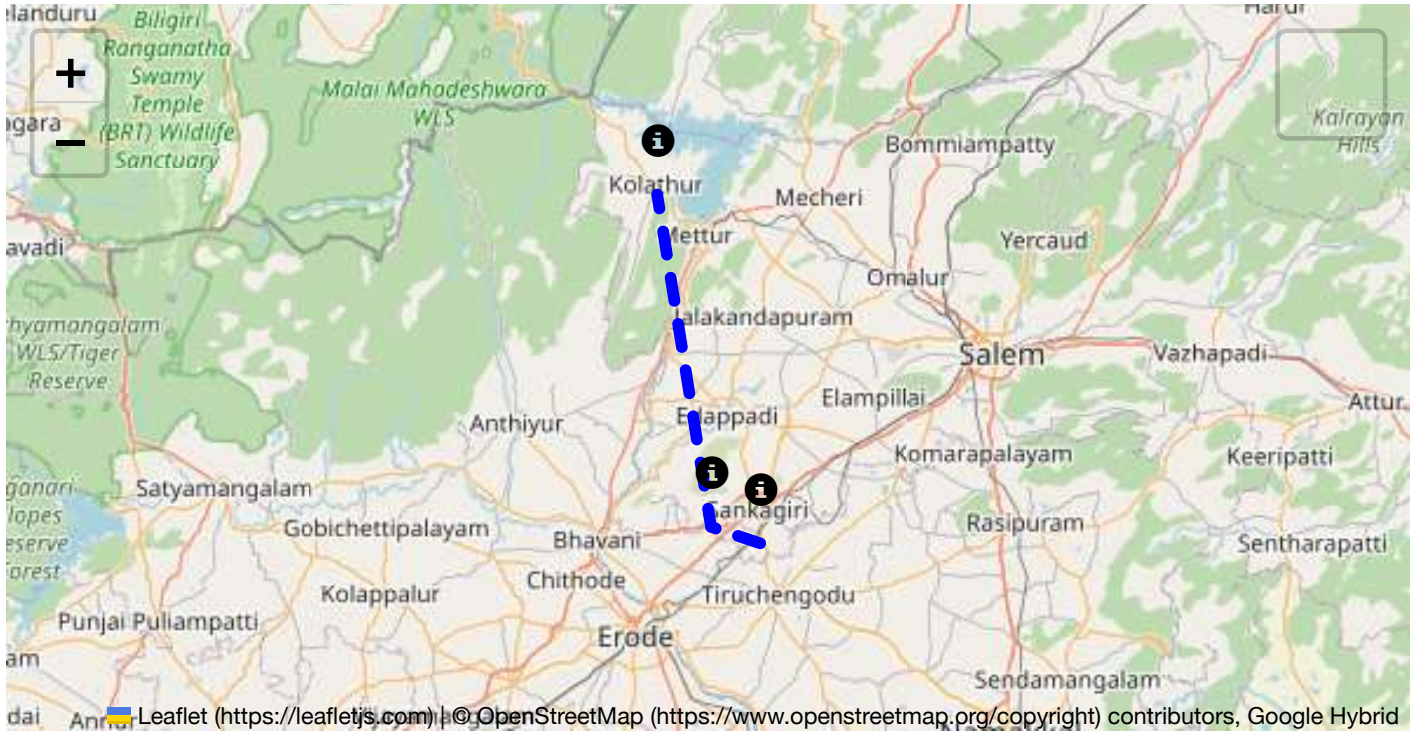
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 69.50 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours

Start: (11.4381, 77.8734)
End: (11.837584, 77.751701)



Welcome to the Journey Risk Management Study

Route Overview

The route spans approximately 69.50 kilometers, starting at CVQF+23W in Sangagiri, Tamil Nadu, passing through FR48+7XQ on Bhavani Main Road in Padaiveedu, and ending at 12/3 C, Chinna Mettur, Tamil Nadu. The route predominantly follows state highways, known for their variable road conditions and traffic density.

Weather Conditions and Potential Hazards

Tamil Nadu typically experiences a tropical climate, with heavy rainfall during the monsoon seasons (June to September and October to December). This can lead to flooded roads and reduced visibility. Summers can be extremely hot, potentially affecting vehicle performance and causing heat-related stress.

Traffic Patterns

Traffic congestion is prevalent during the morning (7:30 AM to 10:00 AM) and evening (5:30 PM to 8:00 PM) peak hours, especially near dense urban settlements like Sangagiri and Mettur. The Bhavani Main Road serves as a critical connector, often seeing high traffic. Congestion can be exacerbated by local festivals and market days.

Road Quality and Infrastructure

While state highways are generally well-maintained, some segments may have potholes or uneven pavement. This is especially true in rural areas, where maintenance may lag. Watch for narrow bridges and sharp turns, particularly around Padaiveedu, which can pose challenges for heavy vehicles.

Alternative Routes for Emergencies

In the event of roadblocks or severe congestion, alternative routes via local roads connecting through Erode or bypassing via Salem may be viable options. However, these routes may add significant travel time and distance.

Local Regulations on Hazardous Material Transport

Transporting hazardous materials through Tamil Nadu requires adherence to stringent regulations. Ensure all necessary permits are secured, and vehicles are clearly marked. There may be restrictions on transporting hazardous materials through certain densely populated areas or during specific hours to minimize public risk.

Historical Incidents

While specific historical data for this exact route is not readily available, Tamil Nadu has recorded incidents involving heavy vehicles, primarily due to overloading or brake failures on hilly terrains. Awareness and adherence to load limits and vehicle checks before departure can mitigate risks.

Environmental Considerations and Sensitive Areas

Sensitive ecological zones may not be prominent along this route; however, rural areas with agricultural activities may require cautious driving to prevent environmental damage. Spillage of hazardous materials could significantly impact local water sources and soil.

Communication Coverage

Though mobile network coverage is generally reliable along state highways, dead zones may exist in rural stretches, particularly near Padaiveedu. Planning for communication outages by having alternative contact methods is advisable.

Emergency Response Times

Emergency response times can vary widely depending on location. Urban areas like Sangagiri and Mettur may have quicker response times (approximately 20-30 minutes), while rural segments might longer (up to 60 minutes). Establish contacts with local emergency services beforehand.

Overall Risk Assessment Summary

The route poses moderate risks primarily due to variable road conditions, traffic congestion, and weather-related challenges. Adequate preparation, including understanding local regulations and potential alternative routes, can minimize risks. Pre-trip vehicle inspections and regular communication checks are recommended to ensure a safe journey. It is crucial to adhere strictly to all safety regulations, bearing in mind the potential environmental impact and public safety concerns associated with transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60019, 77.81968	15 KM/Hr
2	Roundabout	High	11.77576, 77.79789	15 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	High	11.45353, 77.85697	15 KM/Hr
7	Turn	Medium	11.45853, 77.85139	30 KM/Hr
8	Turn	Medium	11.45906, 77.85132	30 KM/Hr
9	Turn	High	11.46307, 77.84941	15 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45636, 77.81678	15 KM/Hr
12	Blind Spot	Blind Spot	11.49397, 77.86740	10 KM/Hr
13	Turn	Medium	11.49441, 77.86690	30 KM/Hr
14	Turn	Medium	11.49971, 77.86468	30 KM/Hr
15	Turn	Medium	11.50885, 77.86037	30 KM/Hr
16	Turn	Medium	11.52519, 77.85535	30 KM/Hr
17	Turn	Medium	11.52744, 77.85582	30 KM/Hr
18	Turn	Medium	11.55049, 77.85513	30 KM/Hr
19	Turn	Medium	11.55460, 77.85624	30 KM/Hr
20	Turn	High	11.57893, 77.85430	15 KM/Hr
21	Turn	Medium	11.60348, 77.83967	30 KM/Hr
22	Turn	Medium	11.60345, 77.83963	30 KM/Hr
23	Turn	High	11.65412, 77.76440	15 KM/Hr
24	Turn	Medium	11.65611, 77.76463	30 KM/Hr
25	Turn	High	11.65628, 77.76450	15 KM/Hr
26	Turn	Medium	11.65710, 77.76477	30 KM/Hr
27	Turn	High	11.65765, 77.76530	15 KM/Hr
28	Turn	High	11.65764, 77.76613	15 KM/Hr
29	Turn	High	11.66104, 77.76762	15 KM/Hr
30	Turn	Medium	11.66091, 77.76851	30 KM/Hr
31	Turn	Medium	11.66125, 77.76902	30 KM/Hr
32	Turn	High	11.66124, 77.76997	15 KM/Hr
33	Turn	Medium	11.66387, 77.77086	30 KM/Hr
34	Turn	High	11.66406, 77.77137	15 KM/Hr
35	Turn	High	11.66861, 77.77229	15 KM/Hr
36	Turn	High	11.66878, 77.77184	15 KM/Hr
37	Turn	Medium	11.66915, 77.77185	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
38	Turn	Medium	11.66958, 77.77211	30 KM/Hr
39	Turn	Medium	11.72550, 77.78365	30 KM/Hr
40	Turn	Medium	11.72637, 77.78419	30 KM/Hr
41	Blind Spot	Blind Spot	11.72749, 77.78413	10 KM/Hr
42	Turn	Medium	11.72743, 77.78528	30 KM/Hr
43	Turn	Medium	11.72792, 77.78585	30 KM/Hr
44	Turn	Medium	11.73949, 77.78816	30 KM/Hr
45	Turn	Medium	11.73973, 77.78843	30 KM/Hr
46	Turn	High	11.73984, 77.78914	15 KM/Hr
47	Turn	Medium	11.75284, 77.79716	30 KM/Hr
48	Turn	High	11.75608, 77.79873	15 KM/Hr
49	Turn	High	11.75682, 77.79842	15 KM/Hr
50	Turn	High	11.75726, 77.79862	15 KM/Hr
51	Turn	Medium	11.75761, 77.80006	30 KM/Hr
52	Turn	Medium	11.76235, 77.80505	30 KM/Hr
53	Turn	High	11.76320, 77.80533	15 KM/Hr
54	Turn	High	11.76409, 77.80422	15 KM/Hr
55	Turn	Medium	11.76449, 77.80422	30 KM/Hr
56	Turn	High	11.77036, 77.80820	15 KM/Hr
57	Turn	High	11.77052, 77.80816	15 KM/Hr
58	Turn	High	11.79781, 77.80274	15 KM/Hr
59	Turn	Medium	11.79809, 77.80211	30 KM/Hr
60	Turn	High	11.80264, 77.79727	15 KM/Hr
61	Turn	High	11.80340, 77.79754	15 KM/Hr
62	Turn	Medium	11.80348, 77.79746	30 KM/Hr
63	Turn	High	11.80672, 77.79707	15 KM/Hr
64	Turn	Medium	11.80794, 77.79254	30 KM/Hr
65	Turn	Medium	11.80880, 77.79162	30 KM/Hr
66	Turn	Medium	11.80918, 77.78995	30 KM/Hr
67	Turn	Medium	11.80852, 77.78611	30 KM/Hr
68	Turn	Medium	11.80507, 77.78128	30 KM/Hr
69	Turn	Medium	11.80742, 77.77646	30 KM/Hr
70	Turn	Medium	11.81053, 77.77322	30 KM/Hr
71	Turn	Medium	11.81408, 77.77270	30 KM/Hr
72	Turn	Medium	11.81463, 77.77229	30 KM/Hr
73	Turn	Medium	11.81868, 77.77212	30 KM/Hr
74	Turn	Medium	11.81929, 77.77233	30 KM/Hr
75	Turn	Medium	11.82085, 77.77158	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
76	Turn	Medium	11.82627, 77.77302	30 KM/Hr
77	Turn	Medium	11.83569, 77.76969	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	S.K.A Hospital	11.7816111, 77.797688	30 km/h	Medium
3	hospital	Siddharth Maternity Centre	11.7874261, 77.8011632	30 km/h	Medium
4	clinic	AV Clinic	11.787386, 77.800135	30 km/h	Medium
5	hospital	Sathya Hospital	11.7864589, 77.8004201	30 km/h	Medium
6	clinic	Saraswathi Poly Clinic	11.7905491, 77.8007345	30 km/h	Medium
7	clinic	Duraisamy Eye Clinic	11.7942636, 77.7995843	30 km/h	Medium
8	clinic	Shiva Siddha Clinic	11.794237, 77.7995531	30 km/h	Medium
9	clinic	Dhanvanthri Clinic	11.7936896, 77.7994075	30 km/h	Medium
11	hospital	Govt Hospital Mettur	11.7959119, 77.7987959	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Govt. Arts College, Mettur	11.7756806, 77.7958483	30 km/h	Medium
10	school	Govt. Hr. Sec. School	11.792309, 77.8006337	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60019, 77.81968



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77576, 77.79789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



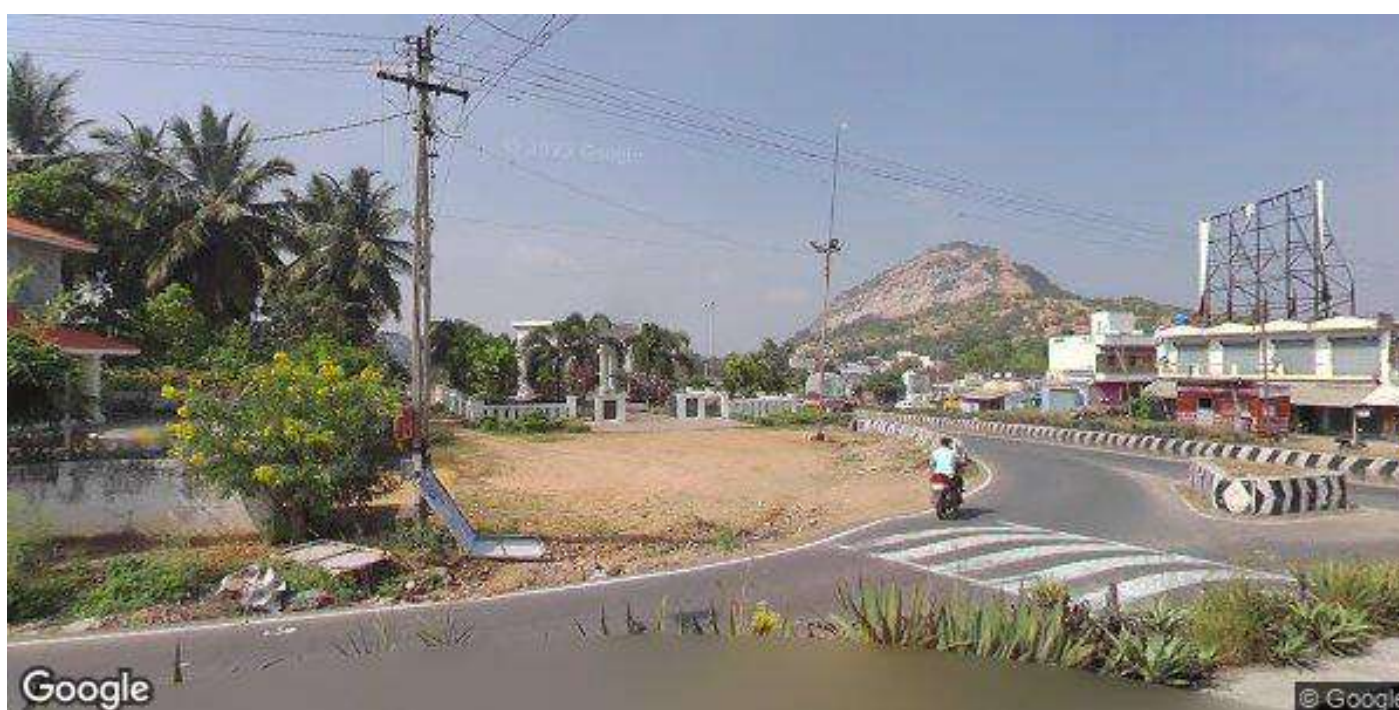
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49397, 77.86740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49441, 77.86690



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49971, 77.86468



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50885, 77.86037



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52519, 77.85535



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52744, 77.85582



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55049, 77.85513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55460, 77.85624



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60348, 77.83967



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60345, 77.83963



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65412, 77.76440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65611, 77.76463



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65628, 77.76450



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65710, 77.76477



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65765, 77.76530



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65764, 77.76613



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66104, 77.76762



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66091, 77.76851



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66125, 77.76902



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66124, 77.76997



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66387, 77.77086



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66406, 77.77137



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66861, 77.77229



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66878, 77.77184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66915, 77.77185



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66958, 77.77211



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72550, 77.78365



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72637, 77.78419



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.72749, 77.78413



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72743, 77.78528



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72792, 77.78585



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73949, 77.78816



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73973, 77.78843



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73984, 77.78914



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75284, 77.79716



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75608, 77.79873



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75682, 77.79842



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75726, 77.79862



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75761, 77.80006



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76235, 77.80505



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76320, 77.80533



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76409, 77.80422



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76449, 77.80422



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77036, 77.80820



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77052, 77.80816



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.79781, 77.80274



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.79809, 77.80211



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.80264, 77.79727



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.80340, 77.79754



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80348, 77.79746



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.80672, 77.79707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80794, 77.79254



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80880, 77.79162



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80918, 77.78995



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80852, 77.78611



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80507, 77.78128



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80742, 77.77646



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81053, 77.77322



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81408, 77.77270



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81463, 77.77229



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81868, 77.77212



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81929, 77.77233



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.82085, 77.77158



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.82627, 77.77302



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83569, 77.76969

